Physics 214/224, Quiz 3

Primary instructor: Yurii Maravin

03/29/2019

For grading purposes

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Studio	
Total:	

WID:__________________________________ NAME:____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

Smith (TU 7:30) Lohman (TU 9:30) Maravin (TU 11:30)

Maravin (TU 1:30) Smith (TU 3:30)

Weliweriya (WF 9:30) Weliweriya (WF 11:30) Smith (WF 1:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.

Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) What is the name of your EP2 professor
   (a) Yurii Maravin
   (b) Brandi Lohman
   (c) Peter Nelson
   (d) Timothy Bolton
   (e) Wait... isn't it a chemistry quiz???

2. (3 points) There is a long wire with a length of 1,000,000,000 m. The current flowing through the wire is 0 A. What is the magnitude of the magnetic field at the distance of 1 m from the wire?
   (a) 0 T
   (b) 1,000,000,000 T
   (c) It depends on the direction of the current
   (d) I like pizza so much!

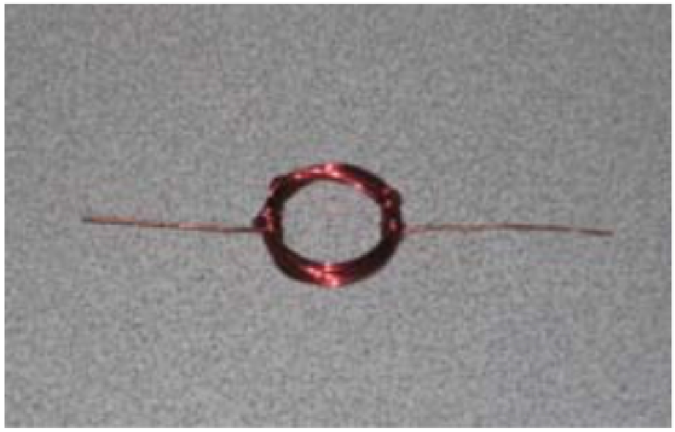
3. (3 points) When applying the right-hand rule, what is often the best extremity to employ?
   (a) right foot
   (b) right hand
   (c) left foot
   (d) left hand

4. (3 points) The needle of a good compass points due north in Kansas. Where would it point in Australia?
   (a) North
   (b) South
   (c) East
   (d) West
   (e) Nowhere, compasses do not work down-under

5. (3 points) Magnetic field lines are
   (a) ... like electric field lines - they start at north pole and terminate at south pole
   (b) ... unlike electric field lines - they form closed loops that pass through north and south pole

6. (3 points) Which current-carrying wire configuration would produce a magnetic field pattern most similar to that produced by the earth?
   (a) A long straight wire
   (b) A toroid
   (c) A solenoid
   (d) A current sheet

7. (3 points) The photo below shows the coil for a DC motor like the one you built in studio. The coil's copper wire is coated with insulation that has to be removed in a particular way to establish a commutator, so that the motor works correctly. What should be done?



(a) No insulation should be removed

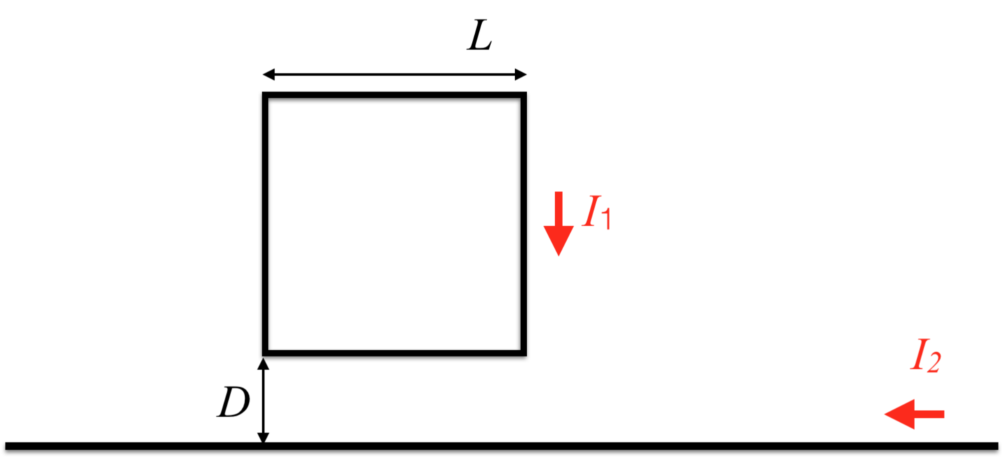
(b) The insulation should be removed completely from both sides

(c) The insulation should be completely removed on the left side, but left completely intact on the right side

(d) The top half of the insulation should be removed from the left side and the bottom half has to be removed from the right side

(e) The top half of the insulation should be removed from the left side and all the insulation has to be removed from the right side

8. (3 points) Which direction will the force produced by the field from the lower wire act on the rectangular current carrying loop?



(a) Away from wire

(b) Towards wire

(c) Out of page

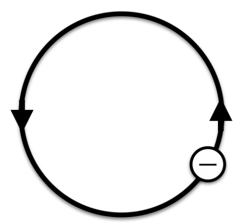
(d) Into the page

(e) Nowhere ($\vec{F} = 0$)

9. (3 points) Suppose you stood directly underneath high voltage DC power lines carrying current towards the south. Which way would the magnetic field produced by these wires point at your position?

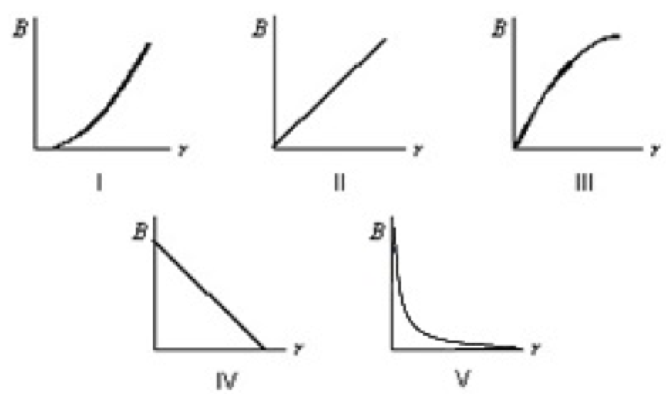
   (a) North
   (b) South
   (c) East
   (d) West
   (e) Nowhere ($\vec{B} = 0$)

10. (3 points) Electrons are going around a circle in a counterclockwise direction as shown. At the center of the circle they produce a magnetic field that is:



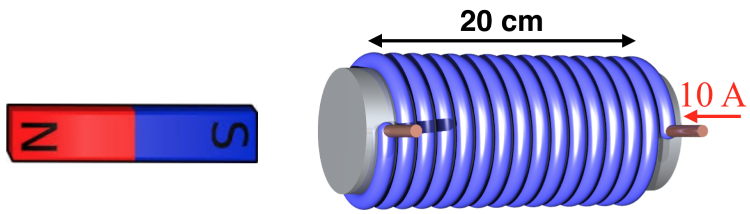
   (a) into the page
   (b) out of the page
   (c) to the left
   (d) to the right
   (e) zero

11. (3 points) Which graph shown below correctly gives the magnitude of the magnetic field outside an infinitely long, very thin, straight current-carrying wire as a function of the distance r from the wire?



   (a) I
   (b) II
   (c) III
   (d) IV
   (e) V

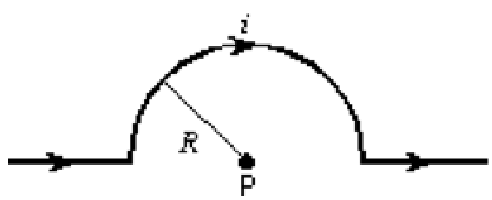
The next two problems refer to the solenoid described below, with current flowing in the direction shown.



12. (3 points) The magnetic field at the center of the solenoid has a magnitude
   (a) $B = 1.0 \times 10^{-4}$ T
   (b) $B = 9.4 \times 10^{-4}$ T
   (c) $B = 1.9 \times 10^{-3}$ T
   (d) $B = 3.8 \times 10^{-2}$ T

13. (3 points) The bar magnet to the left of the solenoid in the picture above will be
   (a) Rotated clockwise
   (b) Pushed left
   (c) Rotated counter-clockwise
   (d) Pulled right
   (e) Stay the way it is

14. (3 point) The magnitude of the magnetic field at point P at the center of the semicircle shown is given by



   (a) $2\mu_0 i/R^2$
   (b) $2\mu_0 i/2\pi R$
   (c) $\mu_0 i/4\pi R$
   (d) $\mu_0 i/2R$
   (e) $\mu_0 i/4R$

15. (3 points) A piece of aluminum develops a small magnetic field pointing away from the north pole of the bar magnet when the bar magnet is brought close. The field goes away when the bar magnet is removed. Aluminum is
   (a) non-magnetic
   (b) diamagnetic
   (c) paramagnetic
   (d) ferromagnetic

16. (3 points) A piece of unknown material is strongly attracted to a bar magnet and retains magnetization when the bar is removed. The unknown material should have an appreciable amount of

    (a) copper
    (b) aluminum
    (c) water
    (d) iron
    (e) Play-Doh

Part II: Work out this part in the space provided. Show your work! (52 points total)

17. (12 points) Here is a concept for an electromagnetic projectile launcher sometimes called a rail gun. The rail gun consists of a battery supplying a voltage V_0 when switched into the circuit. Two very small resistance parallel conducting rails connect to the battery. The circuit is completed by a bar of length L, mass m, and resistance R that can slide along the rails; the bar serves as the projectile. A magnet produces a field of magnitude B_0 that exerts a force on a sliding bar. Suppose the rails are oriented vertically with the horizontal sliding bar positioned in an east-west direction, and the current in the bar moving towards east when the circuit is connected.

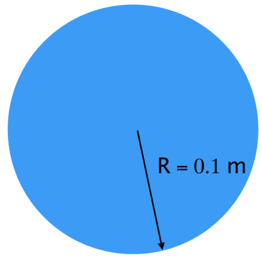
    (a) (3 points) Which direction does the B-field need to point for an upward vertical force to develop on the projectile?

    (b) (6 points) Show that the magnitude of the force on the bar just after the circuit is connected, in terms of V_0, R, B_0, and L, is $F = V_0 B_0 L/R$.

    (c) (3 points) Take $V_0 = 100$ V, $B_0 = 0.1$ T, $L = 10$ cm, $R = 0.1\ \Omega$, and $m = 10$ g and determine the acceleration experienced by the bar.

Extra paper for calculations

18. (12 points) The figure below show a cross section of a long cylindrical conductor of radius $R = 0.1$ m made of pure copper for a groundbreaking experiment "levitate a human in a magnetic field" experiment. A current of 100 A flows through the conductor, uniformly. **In writing down your solution, you need explain your reasoning in words to get the full credit**.



(a) (2 points) What is the magnitude of the magnetic field at the center of the wire ($r = 0$ m)?

(b) (6 points) Find the magnitude of the magnetic field inside the conductor, at a given distance r from the center of the conductor (thus, $r < R$) and evaluate it for $r = 0.05$ m.

(c) (4 points) Find the magnitude of the magnetic field outside the conductor $r \geq R$, and evaluate it at $r = 0.1$ m.

Extra paper for calculations

19. (14 points) Suppose we wish to make an 20 cm long solenoid with a cross-sectional area of 10 cm^2. We have available a 100 m long roll of wire characterized by a resistance-per-length $\frac{dR}{dL} = 0.005\ \Omega/\mathrm{m}$. To operate safely, the solenoid cannot dissipate more than $P_{max} = 100$ W through the heating of its coil. If we used all the available wire and operated the solenoid with $I = 10$ A of current, then:

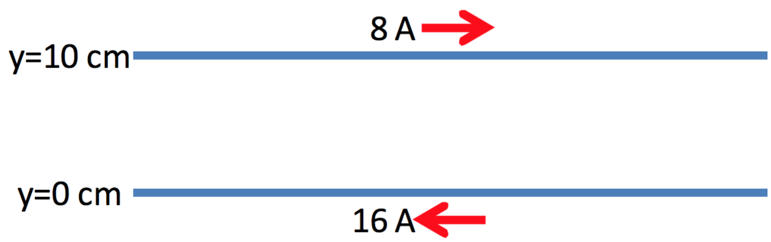
 (a) (4 points) Show that the solenoid will have about $N = 892$ total turns

 (b) (6 points) What would the field be at the center of the solenoid?

 (c) (4 points) Determine whether the coil is operating safely with this current.

Extra paper for calculations

20. (14 points) Two long wires carry the currents shown in the figure below



(a) (4 points) Show that the magnetic field produced by the lower wire at the position of the upper wire has a magnitude of 3.2×10^{-5} T.

(b) (5 points) Determine the magnitude and direction of the magnetic force per unit length on the upper wire.

(c) (5 points) Determine the value of positive y where the net magnetic field will be zero.